

UNIT - 1

2 Mark Questions:

Q1. Define Machine Learning.

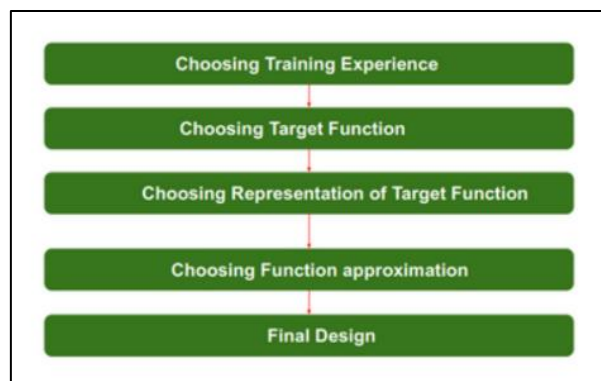
Machine Learning is a branch of artificial intelligence that allows computer systems to automatically learn from data, experiences, and past outcomes without being explicitly programmed. The system improves its performance with time as it gains more experience.

Example: A spam filter that learns to detect spam emails based on user labeling history.

Q2. List out the different issues in Machine Learning.

1. **Insufficient or Poor-Quality Data:** Inaccurate, noisy, or incomplete data leads to poor model performance.
 2. **Overfitting and Underfitting:** Overfitting makes the model too specific; underfitting makes it too general.
 3. **Algorithm Selection:** Choosing the right algorithm for a given task is often difficult.
 4. **Computational Complexity:** Training large models requires high processing power and time.
 5. **Interpretability of Models:** Some models, especially deep learning, act as “black boxes,” making them hard to explain or trust.
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Q3. Draw the diagram for showing the steps involved in designing a learning system.



Q4. Enlist the Advantages and Disadvantages of Unsupervised Learning.

Advantages:

1. Helps identify unknown patterns or structures in data.

2. Useful for clustering and grouping similar data points.
3. Works well with large and complex datasets.

Disadvantages:

1. No predefined output makes evaluation difficult.
 2. Results may be less accurate than supervised learning.
 3. Interpretation of clusters can be challenging.
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Q5. State the Well-Posed Learning Problems with Examples.

A problem is considered well-posed in Machine Learning if it is defined by three components: Task (T), Performance Measure (P), and Experience (E). For example, in email spam detection —

T: Classifying emails as spam or not spam,

P: Accuracy of classification,

E: Collection of labeled emails used for training.

Q6. Write a Short Note on Reinforcement Learning.

Reinforcement Learning is a type of learning where an agent interacts with its environment, makes decisions, and learns from the feedback it receives in the form of rewards or penalties. The main goal is to maximize cumulative rewards through trial and error.

Example: A self-driving car learning to navigate traffic through continuous feedback.

Q7. Give the Advantages and Disadvantages of Supervised Learning.

Advantages:

1. Produces accurate and reliable results.
2. Easy to measure performance using known labels.
3. Useful for solving real-world predictive problems.

Disadvantages:

1. Requires a large amount of labeled training data.
 2. Can struggle with unseen or noisy data.
 3. Model training can be time-consuming and costly.
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Q8. Rewrite the Concept of Semi-Supervised Learning.

Semi-supervised Learning combines the benefits of both supervised and unsupervised learning. It uses a small amount of labeled data along with a large amount of unlabeled data to improve accuracy and reduce training costs.

Example: Image classification where only a few images are labeled but the model learns from the entire dataset.

4 Marks Questions:

Q1. Distinguish between Supervised and Unsupervised Learning.

Basis	Supervised Learning	Unsupervised Learning
Data Used	Uses labeled data with known outputs.	Uses unlabeled data without predefined outputs.
Objective	Learn a mapping from inputs to outputs.	Discover hidden patterns or structure in data.
Feedback	Learns with direct feedback from known results.	No feedback; model identifies structure by itself.
Applications	Classification and Regression problems.	Clustering and Association rule mining.
Accuracy	Usually more accurate due to labeled data.	Often less accurate since data lacks labels.
Examples	Email spam detection, stock price prediction.	Customer segmentation, topic modeling.

Q2. Examine Semi-Supervised Learning with Example, Advantages and Disadvantages.

Semi-Supervised Learning is a hybrid approach that combines a small amount of labeled data with a large amount of unlabeled data to improve model performance. It is useful when labeling data is expensive or time-consuming.

Example: In image recognition, only a few images are labeled (like “cat” or “dog”), while most are unlabeled. The model uses both to learn efficiently.

Advantages:

1. Reduces the cost and time of labeling data.
2. Utilizes large amounts of unlabeled data to enhance accuracy.

3. Provides better performance than purely unsupervised learning.

Disadvantages:

1. The model may learn incorrect patterns from noisy unlabeled data.
 2. Performance heavily depends on the quality of the small labeled dataset.
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Q3. Express the Concept of Well-Posed Learning Problems with Four Examples.

A **well-posed learning problem**, as defined by Tom Mitchell, states that a computer program is said to learn from experience (E) concerning some task (T) and performance measure (P) if its performance on T, as measured by P, improves with E.

It ensures that the learning process is measurable and repeatable.

Examples:

- | | |
|--|---|
| ○ Spam Detection:
T – Classify emails as spam or not spam
P – Accuracy of classification
E – Labeled emails. | ○ Handwriting Recognition:
T – Identify handwritten words
P – Percent correctly recognized
E – Labeled handwriting dataset. |
| ○ Checkers Game:
T – Playing checkers
P – Percentage of games won
E – Experience from self-play. | ○ Self-driving Car:
T – Drive safely
P – Distance traveled without errors
E – Observations of human driving. |
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Q4. Illustrate and List the Different Issues in Machine Learning.

Machine Learning faces several key issues that affect the success and reliability of models.

1. **Data Quality and Quantity:** Insufficient or noisy data leads to poor model generalization.
2. **Overfitting and Underfitting:** Overfitting occurs when the model memorizes data; underfitting happens when it fails to learn the trend.
3. **Selection of Algorithms:** Choosing the right model for a given task is often complex.
4. **Computational Complexity:** Training large models requires high computational power and time.
5. **Interpretability:** Some models, like deep learning, are difficult to interpret or explain.
6. **Bias and Ethics:** Improper data collection may introduce unfair or biased predictions.

Example: In loan approval systems, biased data can lead to unfair rejections of applicants.

7 Mark Questions:

Q1. Explain the Different Applications of Machine Learning.

Machine Learning (ML) is widely applied across various domains where systems can learn from data to make predictions or decisions without explicit programming. It helps automate complex tasks and uncover patterns in large datasets.

Applications of Machine Learning:

1. **Email Spam Detection:** ML algorithms such as Naïve Bayes classify emails as spam or not spam by learning from labeled email datasets.
 2. **Image and Face Recognition:** Systems like Google Photos or facial authentication use ML to recognize faces and objects in images.
 3. **Speech Recognition:** Virtual assistants like Siri and Alexa use ML to understand and convert spoken language into text.
 4. **Medical Diagnosis:** ML helps doctors detect diseases like cancer or diabetes by analyzing medical images and patient records.
 5. **Recommendation Systems:** E-commerce sites and OTT platforms use ML to recommend products or movies based on user behavior.
 6. **Financial Fraud Detection:** ML models monitor transaction patterns to detect unusual or fraudulent activities in banking systems.
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Q2. Examine the Steps in Developing a Machine Learning System.

Developing a Machine Learning system involves several well-defined steps that transform raw data into a trained model capable of making accurate predictions.

Steps in Developing a Machine Learning System:

1. **Choosing the Training Experience:** Select meaningful training data that directly or indirectly improves the model's performance.
2. **Choosing the Target Function:** Define what the system should learn — for example, a function that predicts whether an email is spam.
3. **Choosing Representation for Target Function:** Represent the target function using mathematical or logical forms such as tables, graphs, or linear equations.
4. **Choosing a Function Approximation Algorithm:** Select an algorithm (like decision trees or regression) to approximate the target function from data.

5. **Training the Model:** Feed the algorithm with data to help it learn the underlying pattern or relationship.
 6. **Evaluating the Model:** Test the trained model on new data to check its performance and accuracy.
 7. **Final Design and Deployment:** Optimize and deploy the model for real-world use, improving it continuously with new experience.
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Q3. Illustrate the Various Real-World Applications of Machine Learning.

Machine Learning has become a core part of modern technology, enabling systems to learn automatically and improve from experience. It is widely used in many real-world scenarios.

Real-World Applications:

1. **Healthcare:** ML assists in diagnosing diseases, predicting patient outcomes, and personalizing treatment plans.
Example: Predicting heart disease based on patient health records.
 2. **Finance:** Used in credit scoring, stock price prediction, and fraud detection by analyzing transaction patterns.
Example: Banks use ML to detect suspicious credit card activity.*
 3. **Transportation:** ML powers self-driving cars, traffic prediction, and route optimization.
Example: Google Maps predicts traffic congestion using ML.
 4. **Retail and E-commerce:** Recommender systems suggest products based on user history and preferences.
Example: Amazon's product recommendations use ML algorithms.
 5. **Social Media:** Platforms use ML for content recommendations, facial recognition, and fake account detection.
Example: Facebook auto-tags friends in uploaded photos.*
 6. **Agriculture:** ML models help monitor crop health, predict yields, and detect pests using satellite images.
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Q4. Describe the complete Life Cycle of Machine Learning by describing each phase in detail.

The **Machine Learning Life Cycle** defines the step-by-step process of building, training, and deploying a learning model. Each phase ensures the system learns effectively from data and provides accurate results.

Phases of Machine Learning Life Cycle:

1. Data Collection:

Gather relevant and high-quality data from multiple sources. The quality of input data directly affects model accuracy.

Example: Collecting historical sales data to predict future sales.

2. Data Preprocessing:

Clean, transform, and organize data by handling missing values, removing duplicates, and normalizing data for consistency.

3. Model Selection:

Choose a suitable algorithm (e.g., Decision Tree, Regression, or K-Means) based on the problem type — classification, regression, or clustering.

4. Model Training:

Feed the preprocessed data to the chosen algorithm so it can learn patterns and relationships.

5. Model Evaluation:

Test the trained model using separate test data and evaluate it using performance metrics like accuracy or error rate.

6. Deployment and Monitoring:

Deploy the model for real-world use and continuously monitor its performance to make improvements.

Q5. Analyse the Advantages & Disadvantages of the following:

a) Supervised Learning

Advantages:

1. Produces accurate and reliable results.
2. Easy to measure performance using labeled data.
3. Suitable for both classification and regression problems.

Disadvantages:

1. Requires large labeled datasets.
2. May not handle unseen data well.
3. Training can be costly and time-consuming.

Example: Email spam detection.

b) Unsupervised Learning

Advantages:

1. Can find hidden patterns in unlabeled data.
2. Useful for clustering and segmentation tasks.
3. Works with large and complex data.

Disadvantages:

1. No clear evaluation method.
2. May give less accurate results.
3. Hard to interpret outcomes.

Example: Market segmentation.

c) Reinforcement Learning

Advantages:

1. Learns from interaction with environment.
2. Suitable for sequential decision-making tasks.
3. Can achieve high performance through rewards.

Disadvantages:

1. Requires many iterations to learn.
2. High computational cost.
3. Difficult to design proper reward functions.

Example: Self-driving cars.

d) Semi-Supervised Learning

Advantages:

1. Combines benefits of labeled and unlabeled data.
2. Reduces cost and effort of manual labeling.
3. Achieves better accuracy than unsupervised learning.

Disadvantages:

1. Sensitive to data quality.
2. May misclassify if unlabeled data are noisy.
3. Limited use for very small datasets.

Example: Image classification with few labeled samples.

Q6. Examine the steps involved in designing a learning system with showing the diagram.

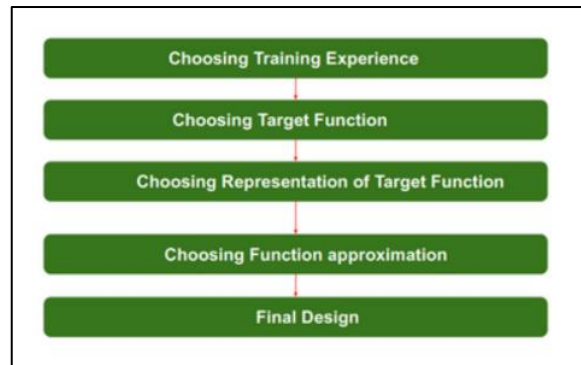
Designing a learning system involves several systematic steps that guide how a computer learns from experience to perform a specific task efficiently.

Steps in Designing a Learning System:

1. Choosing the Training Experience:

Select meaningful and relevant training data or experience that directly impacts model success.

Example: In a chess-playing program, training experience comes from past games.



2. Choosing the Target Function:

Define the goal or output of the system — what it needs to predict or learn.

Example: Function `NextMove(board)` determines the best move in chess.

3. Choosing Representation for Target Function:

Represent the target function in a way that algorithms can process — such as decision trees, graphs, or equations.

4. Choosing Function Approximation Algorithm:

Select an algorithm (e.g., ID3, linear regression) that best fits the problem and approximates the target function.

5. Final Design of the System:

Combine all components, train the model through repeated experience, and evaluate its performance to refine learning.

12 Mark Questions:

Q1. Describe the Following About Machine Learning:

(a) Applications of Machine Learning

Machine Learning has wide-ranging applications in both research and industry. Some of the major areas include:

1. Email Spam Detection:

ML algorithms like Naïve Bayes analyze email content to classify messages as *spam* or *not spam* based on previous labeled data.

Example: Gmail automatically filters spam messages from the inbox.

2. Image and Facial Recognition:

ML models identify people, objects, and scenes in images. This is widely used in social media tagging and security systems.

Example: Face unlock feature in smartphones.

3. **Speech Recognition:**

ML enables voice-based systems like Siri, Alexa, and Google Assistant to understand and process human speech.

4. **Medical Diagnosis:**

ML aids in early disease detection, image-based diagnosis, and treatment recommendations.

Example: Predicting cancer or diabetes using patient data.

5. **Recommendation Systems:**

Online platforms such as Netflix, Amazon, and YouTube use ML to suggest products or content based on user preferences.

6. **Financial Fraud Detection:**

ML algorithms detect unusual transaction patterns and alert banks about potential fraudulent activities.

(b) Life Cycle of Machine Learning

1. **Data Collection:**

Gather high-quality, relevant data from multiple sources. The dataset must represent real-world conditions for effective learning.

Example: Collecting customer data for sales prediction.

2. **Data Preprocessing:**

Clean and organize data by handling missing values, duplicates, and normalization. This ensures better model accuracy.

3. **Model Selection:**

Choose a suitable algorithm based on problem type — e.g., Decision Tree for classification or Linear Regression for prediction.

4. **Model Training:**

Feed the processed data to the algorithm so that it can learn the relationship between inputs and outputs.

5. **Model Evaluation:**

Test the model on unseen data to check its accuracy, precision, and error rate using metrics such as RMSE or confusion matrix.

6. **Model Deployment:**

Deploy the trained model in a real-world environment, where it begins making predictions or automating tasks.

7. Monitoring and Maintenance:

Continuously track model performance, update it with new data, and improve its accuracy over time.

Q2. Analyze the Concept of Machine Learning and Evaluate its Different Types—Supervised, Unsupervised, Semi-Supervised and Reinforcement Learning—with Suitable Real-World Examples.

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Q.3 Explain the concept of Machine Learning and evaluate its different types with suitable real-world examples.

Machine Learning (ML) is a subfield of Artificial Intelligence that enables systems to automatically learn and improve from experience without being explicitly programmed. It focuses on developing algorithms that can analyze data, identify patterns, and make intelligent decisions.

Machine Learning models are broadly classified into **Supervised, Unsupervised, Semi-Supervised, and Reinforcement Learning**—each serving different purposes depending on the data and the problem type.

1. Supervised Learning:

In Supervised Learning, the algorithm is trained using **labeled data**, meaning both input and output values are known. The goal is to learn a mapping function that predicts outputs for new, unseen inputs.

Applications: Classification and regression problems.

Examples:

- Email spam detection
- Predicting house prices using past data.

Advantages: High accuracy, measurable performance.

Disadvantages: Requires large labeled datasets and may fail with unseen data.

2. Unsupervised Learning:

Unsupervised Learning deals with **unlabeled data**, where the model identifies hidden patterns or structures without predefined outputs.

Applications: Clustering and association rule mining.

Examples:

- Customer segmentation in marketing
- Grouping news articles by topic.

Advantages: Finds unknown relationships in data.

Disadvantages: Difficult to evaluate and interpret results.

3. Semi-Supervised Learning:

Semi-Supervised Learning is a **combination of supervised and unsupervised** techniques. It uses a small amount of labeled data and a large amount of unlabeled data to train models efficiently.

Applications: Useful when labeling data is costly or time-consuming.

Examples:

- Image recognition where few images are labeled and many are not
- Speech recognition systems.

Advantages: Reduces labeling effort while maintaining good accuracy.

Disadvantages: Sensitive to data quality; may misclassify noisy data.

4. Reinforcement Learning:

Reinforcement Learning is based on **trial and error**. The system (agent) interacts with its environment and learns by receiving feedback in the form of rewards or penalties.

Applications: Used in dynamic decision-making tasks.

Examples:

- Self-driving cars learning to navigate traffic.
- Game-playing bots like AlphaGo

Advantages: Suitable for complex control problems; learns optimal strategies.

Disadvantages: Requires many iterations and high computational resources.

UNIT - 2

2 Mark Questions:

Q1. Identify the Concept of K-Means Clustering.

K-Means Clustering is an **unsupervised learning algorithm** that groups unlabeled data into *K distinct clusters* based on similarity. Each cluster is represented by a **centroid**, and the algorithm minimizes the distance between points and their cluster centroid.

Example: Grouping customers based on purchasing habits.

Q2. Illustrate the Applications of Clustering.

Clustering is used to **discover patterns and group similar data** in various fields. It is applied in **market segmentation, image compression, document classification, customer grouping, and disease pattern analysis.**

Example: Grouping news articles by topic using text similarity.

Q3. Identify the Concept of Hierarchical Clustering.

Hierarchical Clustering is an **unsupervised learning method** that groups similar data points by forming a **tree-like structure** (dendrogram). It can be **agglomerative (bottom-up)** or **divisive (top-down)** in approach.

Example: Grouping animals based on genetic similarity.

Q4. Classify the Different Applications of Clustering.

Clustering applications can be classified as:

1. **Business Applications:** Customer segmentation, product categorization.
2. **Scientific Applications:** Gene expression analysis, weather data grouping.
3. **Image and Text Processing:** Image compression, document grouping.

Example: Clustering social media users based on interests.

Q5. State the Term Hierarchical Clustering with Example.

Hierarchical Clustering is a technique that **builds a hierarchy of clusters** by continuously merging or splitting data based on similarity. The result is shown using a **dendrogram**.

Example: Merging fruits by weight — grape and cherry combine first, then apple and banana.

Q6. Write a Short Note on Clustering.

Clustering is an **unsupervised learning process** that groups unlabeled data into clusters such that points in the same cluster are similar and different from other clusters. It helps reveal hidden patterns in data.

Example: Dividing customers into groups based on shopping behavior.

4 Mark Questions

Q1. Write a Short Note on K-Means Clustering.

K-Means Clustering is an **unsupervised learning algorithm** used to group unlabeled data into *K distinct clusters* based on similarity. It is a **centroid-based** algorithm, where each cluster is represented by a central point called a **centroid**. The main goal is to minimize the distance between data points and their assigned cluster centroids.

Steps in K-Means:

1. Select the number of clusters (K).
2. Choose random K centroids.
3. Assign each data point to the nearest centroid.
4. Recalculate centroids and repeat until stable clusters form.

Example: Grouping customers based on spending patterns in a shopping mall.

Q2. Examine Different Applications of Clustering.

Clustering has a wide range of applications across various domains as it helps uncover hidden patterns in data.

1. Business and Marketing: Used for customer segmentation, identifying purchasing trends, and targeted advertising.

2. Healthcare: Helps detect disease patterns and group patients with similar symptoms.

3. Image Processing: Used in image compression and object recognition by grouping pixels with similar color intensity.

4. Text and Web Mining: Groups similar documents or webpages for efficient information retrieval.

5. Social Media Analysis: Used to detect communities and analyze user behavior.

Example: Netflix uses clustering to group viewers with similar viewing preferences for better recommendations.

Q3. Express the Step-by-Step Algorithm of Batchelor–Wilkins.

The **Batchelor–Wilkins algorithm** is used to determine the **optimal number of clusters (K)** for clustering algorithms like K-Means. It solves the problem of selecting natural groupings in a dataset.

Steps:

1. **Initial Clustering:** Perform clustering (e.g., K-Means) for different K values such as 2, 3, 4, etc.
2. **Evaluate Cluster Validity:** For each K, calculate evaluation indices like *Silhouette Score*, *Davies–Bouldin Index*, or *Calinski–Harabasz Index*.
3. **Select the Best K:** Choose the K value that gives the best score — indicating compact and well-separated clusters.
4. **Finalize Clusters:** With the chosen K, identify cluster centroids and assign each point to the nearest cluster.

Example: Used in customer segmentation to find the most meaningful number of customer groups.

7 Marks Questions:

Q1. Explain the Concept of Hierarchical Agglomerative Clustering (HAC) and Illustrate its Workflow with a Diagram.

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Q.3 Explain Hierarchical Agglomerative Clustering (HAC) and also Illustrate the working process of Hierarchical Agglomerative Clustering (HAC) with the help of a suitable diagram.

Introduction:

Hierarchical Agglomerative Clustering (HAC) is an **unsupervised learning technique** used to group similar data points into clusters. It is a **bottom-up approach** where each data point starts as its own cluster, and pairs of clusters are successively merged based on their similarity until only one large cluster remains. Unlike K-Means, HAC does not require specifying the number of clusters beforehand.

Workflow of Hierarchical Agglomerative Clustering:

1. **Start with Individual Points:**
Each data point is initially treated as a separate cluster.

2. Calculate Distances:

Compute the distance or similarity between all pairs of clusters using measures like Euclidean or Manhattan distance.

3. Merge Closest Clusters:

Identify the two clusters with the smallest distance and merge them into one cluster.

4. Update the Distance Matrix:

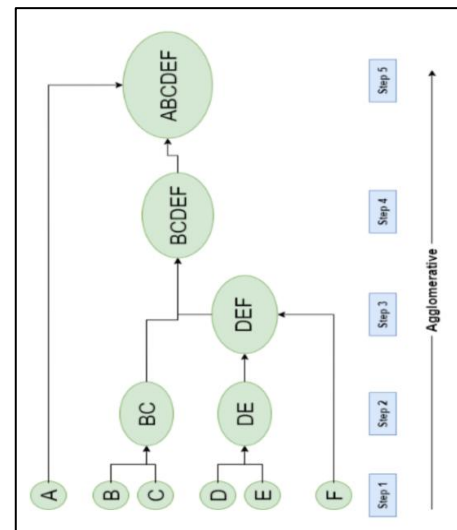
Recalculate distances between the new cluster and all remaining clusters.

5. Repeat the Process:

Continue merging clusters until only a single cluster containing all data points remains.

6. Create a Dendrogram:

The merging process is visualized using a **tree-like diagram** called a **dendrogram**, which shows how clusters are combined step by step.



Example:

Consider fruits with weights: apple (100g), banana (120g), cherry (50g), and grape (30g).

- Grape and cherry merge first (most similar).
- Then apple and banana merge.
- Finally, both clusters merge into one, forming a complete hierarchy.

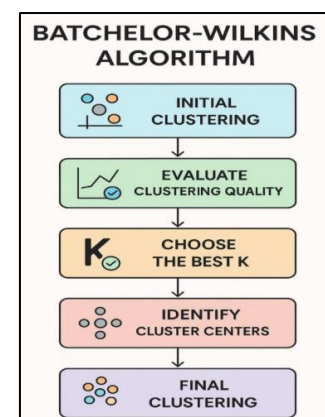
Q2. Analyse about the Following Algorithms: a) Batchelor–Wilkins Algorithm, b) K-Means Clustering Algorithm

(a) Batchelor–Wilkins Algorithm:

The **Batchelor–Wilkins algorithm** addresses the problem of choosing the best number of clusters (K) in a dataset, which is crucial in K-Means clustering.

Steps Involved:

1. **Initial Clustering:** Apply a clustering algorithm (e.g., K-Means) multiple times using different K values (e.g., 2, 3, 4, 5...).
2. **Evaluate Cluster Validity:** For each K, calculate indices such as *Silhouette Score*, *Davies–Bouldin Index*, or *Calinski–Harabasz Index*.
3. **Select Optimal K:** Choose the K that gives the best validity score (high compactness and low separation).



4. **Finalize Clusters:** Perform final clustering with the chosen K and identify the cluster centroids.

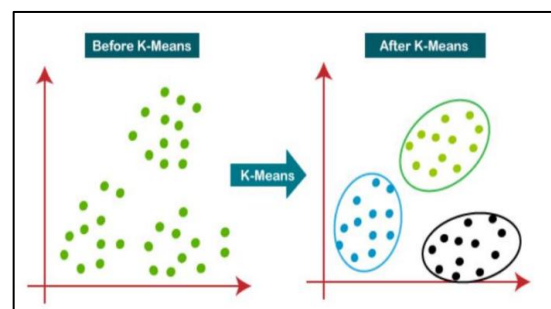
Example: Used in market segmentation to find the most natural number of customer groups.

(b) K-Means Clustering Algorithm

The **K-Means Clustering Algorithm** is one of the most popular **unsupervised learning techniques** used for grouping unlabeled data into K *distinct clusters* based on similarity. It is a **partitional, centroid-based** method that minimizes the total distance between data points and their cluster centroids.

Step-by-Step Working of K-Means:

1. **Select the Number of Clusters (K):**
Decide how many clusters are required in advance.
Example: If $K = 3$, the algorithm will form three clusters.
2. **Initialize Centroids:** Randomly select K points from the dataset as initial cluster centroids.
3. **Assign Data Points to Nearest Centroid:** For each data point, compute the distance (usually Euclidean) to each centroid and assign the point to the nearest one.
4. **Recalculate New Centroids:** After all points are assigned, compute the new centroid by taking the mean of all points in each cluster.
5. **Repeat Iteratively:** Reassign points to the new centroids and update them until there are no changes or minimal movement in centroids. This indicates **convergence**.



Example:

In a retail store, K-Means can group customers into clusters based on purchase frequency and spending patterns — e.g., *low spenders*, *regular shoppers*, and *premium buyers*.

12 Mark Question:

Q1. Rewrite the Following Concepts with Their Algorithms: a) K-Means Clustering, b) Batchelor–Wilkins Algorithm, c) Decision Tree

Machine Learning uses a variety of algorithms to group, classify, and make predictions from data. Among these, **K-Means Clustering** and the **Batchelor–Wilkins Algorithm** belong to *unsupervised learning*, while **Decision Tree** represents a *supervised learning* method. Each algorithm plays a distinct role in organizing or predicting data efficiently.

(a) K-Means Clustering Algorithm

K-Means Clustering is a **centroid-based unsupervised learning algorithm** that divides unlabeled data into K *distinct clusters* based on similarity. Each cluster is represented by a centroid (mean point), and the goal is to minimize intra-cluster distance while maximizing inter-cluster distance.

Algorithm Steps:

1. Choose the number of clusters (**K**) to be created.
2. Randomly select **K centroids** from the dataset.
3. Assign each data point to the nearest centroid based on distance (usually **Euclidean distance**).
4. Recalculate new centroids by finding the mean of all points assigned to each cluster.
5. Repeat steps 3 and 4 until centroids no longer change (i.e., convergence is reached).

Example:

Grouping customers into segments based on purchasing behavior or age and income similarity.

(b) Batchelor–Wilkins Algorithm

The **Batchelor–Wilkins Algorithm** helps determine the **optimal number of clusters (K)** in clustering problems. It evaluates cluster validity to identify the most natural grouping in the data, ensuring better accuracy and meaningful clustering results.

Algorithm Steps:

1. **Initial Clustering:** Run a clustering algorithm (like K-Means) multiple times with different K values (e.g., 2, 3, 4...).
2. **Evaluate Clusters:** For each K , calculate cluster validity indices.
3. **Select Optimal K:** Choose the K value that gives the best validity score (high separation, low compactness).
4. **Finalize Clustering:** Run the clustering algorithm again using the selected K and record final centroids and assignments.

Example:

Used to find the ideal number of customer segments in marketing or grouping documents by topic.

(c) Decision Tree Algorithm

A **Decision Tree** is a **supervised learning algorithm** used for both *classification* and *regression* tasks. It splits data into branches based on feature values, forming a tree-like model where each internal node represents a test on an attribute, each branch represents an outcome, and each leaf node represents a decision or result.

Algorithm Steps:

1. Select the best attribute to split the data using measures like **Information Gain** or **Gini Index**.
2. Divide the dataset into subsets based on the chosen attribute's possible values.
3. Repeat the process recursively for each subset until all data points are classified or the tree reaches a stopping condition.
4. The resulting structure is a **Decision Tree**, which can then classify new instances based on attribute tests.

Example:

Used in credit risk analysis to decide whether a customer should be approved for a loan based on income, credit score, and past behavior.

UNIT - 3

2 Mark Questions:

Q.1 From the deck of Cards, Find the Probability of the card being picked is King Given that it is the Face Card.

- This can be represented as: $p(\text{King} | \text{Face})$.
- There are total 52 cards in a desk ($n=52$), from Which 12 cards are Face Cards: King, Queen and jack with Club, Diamond, heart and Spade.
- There is a set of Face cards with 12 members.
- There is a Subset of King Cards with 4-members. So, 4 Face cards out of 12 Face cards ($4/12$) are King cards. Use Bayes Theorem to Calculate.

Given that,

$n = 52$ King = 4 cards Jack = 4 cards
 Face cards = 12 Queen = 4 cards

According to Bayes theorem

$$P(\text{King} | \text{Face}) = \frac{P(\text{Face} | \text{King}) \cdot P(\text{King})}{P(\text{Face})} \quad \text{--- (1)}$$

→ 4 members are king cards out of 52 cards
 $\therefore P(\text{King}) = \frac{4}{52} = \frac{1}{13}$

→ 12 members are face cards out of 52 cards
 $\therefore P(\text{Face}) = \frac{12}{52} = \frac{3}{13}$

→ Probability of being face given king:

$$P(\text{Face} | \text{King}) = 1$$

As we have prior knowledge if the card is king, it is face card only it is 100% sure so it is always 1.

$\therefore$ Eq. (1) becomes:

$$P(\text{King} | \text{Face}) = \frac{1 \times \frac{1}{13}}{\frac{3}{13}} = \frac{1}{3}$$

$\therefore P(\text{King} | \text{Face}) = \frac{1}{3}$

Q2. Restate the Concept of the Bayes Optimal Classifier and Illustrate it Using Its Formula.

It is a **probabilistic model** that makes the **most probable prediction** for a new example. Basically, it is used to **solve classification problems** using the **probabilistic concept of Bayes' theorem**.

Formula:

$$P(y | x_1, x_2, \dots, x_n) = P(y) \prod_{i=1}^n P(x_i | y)$$

Q3. Restate the Concept of Bayes' Theorem and Illustrate it Using Its Formula.

Bayes' Theorem, also known as **Bayes' Rule** or **Bayesian Rule**, describes the probability of an event based on prior knowledge of conditions that might be related to that event.

It relates the **conditional probability** and **marginal probabilities** of two random events. Bayes' Theorem provides a way to **update or revise probability predictions** of an event by observing new evidence or information related to it.

Formula:

$$P(y|x) = \frac{P(x|y) P(y)}{P(x)}$$

where

$P(y)$: prior probability of class,

$P(x|y)$: likelihood,

$P(y|x)$: posterior probability,

$P(x)$: evidence or normalization constant.

Example: Calculating the probability of having flu given symptoms like fever and cough.

Q4. Identify the Concept of the Bayes Optimal Classifier and Express it Mathematically Using the Appropriate Formula.

It is a **probabilistic model** that makes the **most probable prediction** for a new example. Basically, it is used to **solve classification problems** using the **probabilistic concept of Bayes' theorem**.

Given a dataset $\{x_1, x_2, x_3, \dots, x_n\}$ and $\{y\}$.

$$P(y|x_1, x_2, \dots, x_n) = [P(x_1|y) \cdot P(x_2|y) \cdot \dots \cdot P(x_n|y)] \cdot P(y)$$
$$P(y|x_1, x_2, \dots, x_n) = \frac{\prod_{i=1}^n P(x_i|y) \cdot P(y)}{P(x_1, x_2, \dots, x_n)}$$

Discard the denominator because it is constant

$$P(y|x_1, x_2, \dots, x_n) = P(y) \prod_{i=1}^n P(x_i|y)$$

Example: Predicting a student's performance by averaging probabilities across multiple predictive models.

Q5. Underline the Concept of Naïve Bayes Classifier.

It is a **supervised learning algorithm** based on **Bayes' Theorem** and is mainly used for **solving classification problems**. It is a **probabilistic classifier**, which means it makes predictions based on the **probability of belonging to a particular class**. It is called

“**Naïve**” because it assumes that the **occurrence of each feature is independent** of the others, and “**Baye’s**” because it is based on Baye’s theorem.

Formula:

$$P(A | B) = \frac{P(B | A) \cdot P(A)}{P(B)}$$

Example: Predicting whether an email is spam based on words like “win,” “offer,” and “discount.”

Q.6 In a Certain Population, the Probability that a person has Dengue is $P(D)=1/30000$. Dengue Fever Cusses Neck Pain 80% of the time. The Probability that a person experiences neck pain from any cause is $P(N)=0.02$ what is the probability that a person has dengue given that they have neck pain? Use Bayes’ theorem to calculate.

Handwritten solution for Q.6:

$$P(D) = \frac{1}{30000} \quad P(N|D) = 80\% = \frac{80}{100} = 0.8$$

$$P(N) = 0.02 \quad P(D|N) = ?$$

Using Bayes' Theorem:

$$P(D|N) = \frac{P(N|D) \cdot P(D)}{P(N)}$$

$$P(D|N) = \frac{0.8 \times \frac{1}{30000}}{0.02} = \frac{0.8}{600} = \frac{1}{750} = 0.00133$$

4 Mark Questions:

Q.1 Illustrate the concept of the Bayes Theorem and illustrate it using its formula.

Bayes’ Theorem, also known as **Bayes’ Rule** or **Bayesian Rule**, describes the probability of an event based on prior knowledge of conditions that might be related to that event. It relates the **conditional probability** and **marginal probabilities** of two random events. Bayes’ Theorem provides a way to **update or revise probability predictions** of an event by observing new evidence or information related to it.

Formula:

$$P(y|x) = \frac{P(x|y) P(y)}{P(x)}$$

where

$P(y)$: prior probability of class,

$P(x|y)$: likelihood,

$P(y|x)$: posterior probability,

$P(x)$: evidence or normalization constant.

Example: Calculating the probability of having flu given symptoms like fever and cough.

Q.2 For the Given Dataset, Apply Naïve –Bayes Algorithm and Predict the Outcome for a car, Car= {Red, Domestic, SUV}

color	Type	origin	stolen
Red	Sports	Domestic	yes
Red	Sports	Domestic	No
Red	Sports	Domestic	yes
yellow	Sports	Domestic	No
yellow	Sports	Imported	yes
yellow	SUV	Imported	No
yellow	SUV	Imported	yes
yellow	SUV	Domestic	No
Red	SUV	Imported	No
Red	Sports	Imported	yes

calculate the prior probability for car:

$$P(\text{stolen} = \text{yes}) = \frac{5}{10}$$

$$P(\text{stolen} = \text{No}) = \frac{5}{10}$$

calculate conditional probability for attributes i.e. color, type and origin.

1) For color:

Frequency table	yes	No
color → Red	3	2
→ yellow	2	3

→ calculate the probability for Red=yes and Red=No:

$$P(\text{Red}|\text{yes}) = \frac{P(Y|R) \cdot P(R)}{P(Y)} = \frac{3/5 \times 5/10}{5/10}$$

$$\therefore P(\text{Red}|\text{yes}) = 0.6$$

$$P(\text{Red}|\text{No}) = \frac{P(N|R) \cdot P(R)}{P(N)} = \frac{2/5 \times 5/10}{5/10}$$

$$\therefore P(\text{Red}|\text{No}) = 0.4$$

→ calculate the probability for yellow=yes and yellow=No.

$$P(\text{yellow}|\text{yes}) = \frac{P(Y|Y) \cdot P(Y)}{P(Y)} = \frac{2/5 \times 5/10}{5/10}$$

$$\therefore P(\text{yellow}|\text{yes}) = 0.4$$

$$P(\text{yellow}|\text{No}) = \frac{P(N|Y) \cdot P(Y)}{P(N)} = \frac{3/5 \times 5/10}{5/10}$$

$$\therefore P(\text{yellow}|\text{No}) = 0.6$$

2) For Type:

Frequency table	yes	No
type → sports	4	2
→ SUV	1	3

→ calculate probability of sports=yes and sports=No:

$$P(\text{sports}|\text{yes}) = \frac{P(Y|S) \cdot P(S)}{P(Y)} = \frac{4/5 \times 5/10}{5/10}$$

$$\therefore P(\text{sports}|\text{yes}) = 0.8$$

$$P(\text{sports}|\text{No}) = \frac{P(N|S) \cdot P(S)}{P(N)} = \frac{2/5 \times 5/10}{5/10}$$

$$\therefore P(\text{sports}|\text{No}) = 0.4$$

→ calculate probability of SUV=yes and SUV=No.

$$P(\text{SUV}|\text{yes}) = \frac{P(Y|S) \cdot P(S)}{P(Y)} = \frac{1/5 \times 5/10}{5/10}$$

$$\therefore P(\text{SUV}|\text{yes}) = 0.2$$

$$P(\text{SUV}|\text{No}) = \frac{P(N|S) \cdot P(S)}{P(N)} = \frac{3/5 \times 5/10}{5/10}$$

$$\therefore P(\text{SUV}|\text{No}) = 0.6$$

3) For origin:

Frequency table	yes	No
origin → Domestic	2	3
→ Imported	3	2

→ calculate probability for domestic=yes and domestic=No.

$$P(\text{Domestic}|\text{yes}) = \frac{P(Y|D) \cdot P(D)}{P(Y)} = \frac{2/5 \times 5/10}{5/10}$$

$$\therefore P(\text{Domestic}|\text{yes}) = 0.4$$

$$P(\text{Domestic}|\text{No}) = \frac{P(N|D) \cdot P(D)}{P(N)} = \frac{3/5 \times 5/10}{5/10}$$

$$\therefore P(\text{Domestic}|\text{No}) = 0.6$$

→ calculate probability for Imported=yes and Imported=No.

$$P(\text{Imported}|\text{yes}) = \frac{P(Y|I) \cdot P(I)}{P(Y)} = \frac{3/5 \times 5/10}{5/10}$$

$$\therefore P(\text{Imported}|\text{yes}) = 0.6$$

$$P(\text{Imported}|\text{No}) = \frac{P(N|I) \cdot P(I)}{P(N)} = \frac{2/5 \times 5/10}{5/10}$$

$$\therefore P(\text{Imported}|\text{No}) = 0.4$$

Car = {Red, SUV, Domestic}

$$P(\text{Car}|\text{yes}) = P(\text{yes}) \times P(\text{Red}|\text{yes}) \times P(\text{SUV}|\text{yes}) \times P(\text{Domestic}|\text{yes})$$

$$= \frac{5}{10} \times 0.6 \times 0.2 \times 0.4$$

$$= 0.024$$

$$P(\text{Car}|\text{No}) = P(\text{No}) \times P(\text{Red}|\text{No}) \times P(\text{SUV}|\text{No}) \times P(\text{Domestic}|\text{No})$$

$$= 0.5 \times 0.4 \times 0.6 \times 0.6$$

$$= 0.072$$

compare both probability i.e.

$$P(\text{Car}|\text{yes}) = 0.024 \quad \text{and} \quad P(\text{Car}|\text{No}) = 0.072$$

$P(\text{Car}|\text{No}) = 0.072$ having higher probability.

→ Predict that Car = No for (Red, SUV, Domestic).

Q.3 Write short note on Naïve Bayes Classifier.

The **Naïve Bayes Classifier** is a **supervised learning algorithm** based on the **principle of Bayes' Theorem**. It is widely used to solve **classification problems**, especially in **text classification** tasks that involve high-dimensional data, such as spam detection or sentiment analysis.

It is called “**Naïve**” because it assumes that all the **features** ($x_1, x_2, \dots, x_n$) are **independent of each other** given the class label **y**, and “**Baye’s**” because it is based on Baye’s theorem. Despite this simple assumption, it performs remarkably well in practice.

It is a **probabilistic model**, meaning it predicts the class of an instance based on the **probability** of that instance belonging to each possible class. The class with the **highest posterior probability** is chosen as the prediction.

Formula:

$$P(A | B) = \frac{P(B | A) \cdot P(A)}{P(B)}$$

Q.4 Given the Training data in the table (Buy Computer Data), Predict the Class of the Following New Example Using Bayes Optimal Classifier. Buy Computer = {Age = <=30, income = medium, student = yes, Credit-Rating = fair}

Q. Age	Income	Student	credit Rating	Buy's computer
<30	High	No	Fair	No
<30	High	No	excellent	No
31...40	High	No	Fair	Yes
>40	Medium	No	Fair	Yes
>40	Low	Yes	Fair	Yes
>40	Low	Yes	excellent	No
31...40	Low	Yes	excellent	Yes
<30	Medium	No	Fair	No
<30	Low	Yes	Fair	Yes
>40	Medium	Yes	Fair	Yes
<30	Medium	Yes	excellent	Yes
31...40	Medium	No	excellent	Yes
31...40	High	Yes	Fair	Yes
>40	Medium	No	excellent	No

→ Given the training data in the table (Buy computer data), predict the class of the following new examples using Naïve Bayes classifier and Bayes optimal classifier

Buy = {age < 30, income = medium, student = yes, credit rating = fair}.

⇒ $P(\text{yes}) = \frac{9}{14}$ $P(\text{No}) = \frac{5}{14}$

Name _____ calculate age probability for age for each year and no.

Frequency table	yes	No	$P(Y)$	$P(N)$
age $<= 30$	2	3	$\frac{2}{9}$	$\frac{3}{9}$
$31 \dots 40$	4	0	$\frac{4}{9}$	0
$(Y) > 40$	3	2	$\frac{3}{9}$	$\frac{2}{9}$

1) Year $P(\text{age} < 30 | \text{yes}) = P(Y | X=30) \cdot P(X=30)$

$$\frac{2}{9} \times \frac{2}{9} = (Y) \cdot (X) \cdot P(Y) \cdot P(X)$$

$$\frac{2}{9} \times \frac{2}{9} = \frac{4}{81} = \frac{2}{9} \times \frac{2}{9}$$

$$P(X=30 | \text{yes}) = 0.2222$$

2) Year $P(X=30 | \text{No}) = P(N | X=30) \cdot P(X=30)$

$$P(N) = \frac{3}{9} \times \frac{2}{9}$$

$$P(X=30 | \text{No}) = 0.6$$

Name _____ calculate for Prof income for each yes and No.

Frequency table	yes	No	$P(Y)$	$P(N)$
Income High	2	2	$\frac{4}{9}$	$\frac{2}{9}$
medium	4	2	$\frac{4}{9}$	$\frac{2}{9}$
Low	3	1	$\frac{4}{9}$	$\frac{1}{9}$

1) For $P(\text{medium} | \text{yes}) = P(Y | M) \cdot P(M)$

$$P(Y) = \frac{4}{9}$$

$$P(M | Y) = 0.4444$$

2

3) For $P(\text{Medium}|\text{No}) = \frac{P(\text{NIM}) \cdot P(\text{M})}{P(\text{N})} = \frac{2/6 \times 6/14}{5/14}$
 $P(\text{MIN}) = 0.4$

calculate P of students from each yes and No.

Frequency table	yes	No	P(Y)	P(N)	
student	yes	6	1	6/9	1/5
	No	3	4		

3) $P(S=\text{yes}|\text{yes}) = \frac{P(Y|SY) \cdot P(SY)}{P(Y)} = \frac{6/7 \times 7/14}{9/14}$
 $P(SY|\text{yes}) = 0.6666$

3) $P(S=\text{yes}|\text{No}) = \frac{P(N|SY) \cdot P(SY)}{P(N)} = \frac{2/7 \times 7/14}{5/14}$
 $P(SY|\text{No}) = 0.2$

calculate P of student rating = fair for yes and No

freq. table	yes	No	P(Y)	P(N)	
CR	Excellent	3	3		
	Fair	6	2	6/9	2/5

3) For $P(\text{Fair}|\text{yes}) = \frac{P(Y|F) \cdot P(F)}{P(Y)} = \frac{6/8 \times 8/14}{9/14}$
 $P(F|Y) = 0.6666$

3

2) $P(\text{FIN}) = \frac{P(\text{NIF}) \cdot P(F)}{P(N)} = \frac{2/8 \times 8/14}{5/14}$
 $P(\text{FIN}) = 0.4$

$P(\text{Buy}|\text{yes}) = P(\leq 30|\text{yes}) \times P(\text{Medium}|\text{yes}) \times P(\text{yes}|\text{yes}) \times P(\text{Fair}|\text{yes}) \times P(\text{yes})$
 $= 0.2222 \times 0.4444 \times 0.6666 \times 0.6666$
 $= 0.0438 \times 9 = 0.0281$

$P(\text{Buy}|\text{No}) = P(\leq 30|\text{No}) \times P(\text{Medium}|\text{No}) \times P(\text{yes}|\text{No}) \times P(\text{Fair}|\text{No}) \times P(\text{No})$
 $= 0.6 \times 0.4 \times 0.2 \times 0.4$
 $= 0.0192 \times 5 = 0.00685$

compare $P(\text{Buy}|\text{yes})$ and $P(\text{Buy}|\text{No})$.
 $P(\text{Buy}|\text{yes}) = 0.0281$ & $P(\text{Buy}|\text{No}) = 0.00685$
 $P(\text{Buy}|\text{yes})$ having higher probability.

2) Bayes optimal classifier
 yes student ≤ 30 , medium, yes, fair 3 will buy.

$P(\text{yes}|\dots) = \frac{9 \times 2^1 \times 6^2 \times 6^2 \times 4}{147 \times 9 \times 9 \times 9 \times 9}$
 $= \frac{8}{567} = 0.0141$

$P(\text{No}|\dots) = \frac{5 \times 3 \times 2 \times 1 \times 2}{14 \times 5 \times 5 \times 5 \times 5}$
 $= \frac{12}{1750} = 0.0068$

4

compare $P(\text{yes}|\dots)$ and $P(\text{No}|\dots)$

$P(\text{yes}|\dots) = 0.0141$ & $P(\text{No}|\dots) = 0.0068$

$P(\text{yes}|\dots)$ having higher probability.

→ we can conclude that student Buy's computer = yes for ≤ 30 , medium, yes, fair

5

07 Marks Questions

Q.1 Analyse the Following Concept and illustrate it using its formula. a) Naïve Bayes Classifier, b) Bayes Optimal Classifier

a) Naïve Bayes Classifier

The **Naïve Bayes Classifier** is a **supervised learning algorithm** based on the **principle of Bayes' Theorem**. It is widely used to solve **classification problems**, especially in **text classification** tasks that involve high-dimensional data, such as spam detection or sentiment analysis.

It is called “**Naïve**” because it assumes that all the **features** ($x_1, x_2, \dots, x_n$) are **independent of each other** given the class label y , and “**Baye's**” because it is based on Baye's theorem. Despite this simple assumption, it performs remarkably well in practice.

It is a **probabilistic model**, meaning it predicts the class of an instance based on the **probability** of that instance belonging to each possible class. The class with the **highest posterior probability** is chosen as the prediction.

Formula:

$$P(A | B) = \frac{P(B | A) \cdot P(A)}{P(B)}$$

b) Bayes Optimal Classifier

It is a **probabilistic model** that makes the **most probable prediction** for a new example. Basically, it is used to **solve classification problems** using the **probabilistic concept of Bayes' theorem**.

The **Bayes Optimal Classifier** is the most accurate probabilistic classifier possible, as it considers **all hypotheses** and selects the class with the **highest posterior probability** given input x . It provides the **minimum classification error** theoretically achievable.

Formula:

$$P(y | x_1, x_2 \dots x_n) = P(y) \prod_{i=1}^n P(x_i | y)$$

Q.2 Given the Training data in the table (Buy Computer Data), Predict the Class of the Following New Example Using Bayes Optimal Classifier. Buy Computer = {Age = <=30, income = medium, student = yes, Credit-Rating = fair} (Same as Q. 4 for 4 marks)

Q.3 Consider the given Dataset, Apply Naïve Bayes Algorithm and Predict that is a Fruit has the following Properties then which type of fruit it is. Fruit = {yellow, Sweet, Long}

Fruit	yellow	sweet	long	Total
Mango	350	450	0	650
Banana	400	300	350	1050
Other	50	100	50	150
Total	800	850	400	1200

First we need to calculate the probability of mango in terms of yellow, sweet and long.

$$P(\text{yellow} | \text{Mango}) = \frac{P(\text{Mango} | \text{yellow}) \cdot P(\text{yellow})}{P(\text{Mango})}$$

$$= \frac{350/800 \cdot 800/1200}{650/1200} = \frac{350}{650} = 0.538$$

$$P(\text{sweet} | \text{Mango}) = \frac{P(\text{Mango} | \text{sweet}) \cdot P(\text{sweet})}{P(\text{Mango})}$$

$$= \frac{450/850 \cdot 850/1200}{650/1200} = \frac{450}{650} = 0.692$$

$$P(\text{long} | \text{Mango}) = \frac{P(\text{Mango} | \text{long}) \cdot P(\text{long})}{P(\text{Mango})}$$

$$= \frac{0/400 \cdot 400/1200}{650/1200} = 0$$

$$P(\text{Fruit} | \text{Mango}) = P(\text{yellow} | \text{Mango}) \times P(\text{sweet} | \text{Mango}) \times P(\text{long} | \text{Mango})$$

$$= 0.538 \times 0.692 \times 0 = 0$$

→ This is not a type of fruit i.e. fruit = {yellow, sweet, long}.

Next we need to calculate the probability of Banana in terms of yellow, sweet and long.

$$P(\text{yellow} | \text{Banana}) = \frac{P(\text{Banana} | \text{yellow}) \cdot P(\text{yellow})}{P(\text{Banana})}$$

$$= \frac{400/800 \cdot 800/1200}{1050/1200} = \frac{400}{1050} = 0.381$$

$$P(\text{sweet} | \text{Banana}) = \frac{P(\text{Banana} | \text{sweet}) \cdot P(\text{sweet})}{P(\text{Banana})}$$

$$= \frac{300/850 \cdot 850/1200}{1050/1200} = \frac{300}{1050} = 0.286$$

$$P(\text{long} | \text{Banana}) = \frac{P(\text{Banana} | \text{long}) \cdot P(\text{long})}{P(\text{Banana})}$$

$$= \frac{350/400 \cdot 400/1200}{1050/1200} = \frac{350}{1050} = 0.333$$

$$P(\text{Fruit} | \text{Banana}) = P(\text{yellow} | \text{Banana}) \times P(\text{sweet} | \text{Banana}) \times P(\text{long} | \text{Banana})$$

$$= 0.381 \times 0.286 \times 0.333 = 0.036$$

$$P(\text{Fruit} | \text{Banana}) = P(\text{yellow} | \text{Banana}) \times P(\text{sweet} | \text{Banana}) \times P(\text{long} | \text{Banana})$$

$$= 0.381 \times 0.286 \times 0.333 = 0.036$$

→ It maybe the type of fruit i.e. yellow, long, sweet.

Next, we need to calculate the probability of other in terms of yellow, sweet, long.

$$P(\text{yellow} | \text{Other}) = \frac{P(\text{Other} | \text{yellow}) \cdot P(\text{yellow})}{P(\text{Other})}$$

$$= \frac{50/800 \cdot 800/1200}{150/1200} = \frac{50}{150} = 0.333$$

$$P(\text{sweet} | \text{Other}) = \frac{P(\text{Other} | \text{sweet}) \cdot P(\text{sweet})}{P(\text{Other})}$$

$$= \frac{100/850 \cdot 850/1200}{150/1200} = \frac{100}{150} = 0.666$$

$$P(\text{long} | \text{Other}) = \frac{P(\text{Other} | \text{long}) \cdot P(\text{long})}{P(\text{Other})}$$

$$= \frac{50/400 \cdot 400/1200}{150/1200} = \frac{50}{150} = 0.333$$

$$P(\text{Fruit} | \text{Other}) = P(\text{yellow} | \text{Other}) \times P(\text{sweet} | \text{Other}) \times P(\text{long} | \text{Other})$$

$$= 0.333 \times 0.666 \times 0.333 = 0.074$$

→ It maybe the type of fruit i.e. yellow, long and sweet.

→ Compare all the probabilities:

$$P(\text{Fruit} | \text{Mango}) = 0$$

$$P(\text{Fruit} | \text{Banana}) = 0.036$$

$$P(\text{Fruit} | \text{Other}) = 0.074$$

∴ P(Fruit | Banana) have highest probability compared to all fruits so that banana belongs to category of fruits = {yellow, sweet, long}.

Q.4 Describe the concept of the Bayes Optimal Classifier and illustrate it using its formula and solve the Following Example: Consider the Given Dataset, Apply Bayes Optimal Classifier Algorithm and predict that if a play has Following Properties. Play Tennis = {Outlook = Sunny, Temperature = hot, Humidity= Normal, Wind = Strong}

Day	Outlook	Humidity	Wind	Play Tennis
D1	Sunny	High	Weak	No
D2	Sunny	High	Strong	No
D3	Overcast	High	Weak	Yes
D4	Rain	High	Weak	Yes
D5	Rain	Normal	Weak	Yes
D6	Rain	Normal	Strong	No
D7	Overcast	Normal	Strong	Yes
D8	Sunny	High	Weak	No
D9	Sunny	Normal	Weak	Yes
D10	Rain	Normal	Weak	Yes
D11	Sunny	Normal	Strong	Yes
D12	Overcast	High	Strong	Yes
D13	Overcast	Normal	Weak	Yes
D14	Rain	High	Strong	No

Probability table for outlook				
	yes	NO	P(Y)	P(N)
sunny	2	3	$\frac{2}{9}$	$\frac{3}{5}$
overcast	4	0	$\frac{4}{9}$	0
Rain	3	2	$\frac{3}{9}$	$\frac{2}{5}$
Total	9	5		

Probability table for temperature				
	yes	NO	P(Y)	P(N)
Hot	2	2	$\frac{2}{9}$	$\frac{2}{5}$
Mild	4	2	$\frac{4}{9}$	$\frac{2}{5}$
cool	3	1	$\frac{3}{9}$	$\frac{1}{5}$
Total	9	5		

Probability table for humidity				
	yes	NO	P(Y)	P(N)
High	3	4	$\frac{3}{9}$	$\frac{4}{5}$
Normal	6	1	$\frac{6}{9}$	$\frac{1}{5}$
Total	9	5		

Probability table for wind				
	yes	NO	P(Y)	P(N)
Strong	3	3	$\frac{3}{9}$	$\frac{3}{5}$
weak	6	2	$\frac{6}{9}$	$\frac{2}{5}$
Total	9	5		

Probability table for Play tennis				
	yes	NO		
	9	$\frac{9}{14}$		
	5	$\frac{5}{14}$		
Total	14			

For yes, probability w.r.t {sunny, hot, Normal, Strong}?

$$P(\text{yes} | \text{sunny, hot, Normal, Strong}) = P(\text{sunny} | \text{yes}) \times P(\text{hot} | \text{yes}) \times P(\text{Normal} | \text{yes}) \times P(\text{Strong} | \text{yes}) \times P(\text{yes})$$

$$= \frac{2}{9} \times \frac{2}{9} \times \frac{6}{9} \times \frac{3}{9} \times \frac{9}{14}$$

$$= 0.00705$$

For NO, probability w.r.t {sunny, hot, Normal, Strong}?

$$P(\text{NO} | \text{sunny, hot, Normal, Strong}) = P(\text{sunny} | \text{NO}) \times P(\text{hot} | \text{NO}) \times P(\text{Strong} | \text{NO}) \times P(\text{Normal} | \text{NO}) \times P(\text{NO})$$

$$= \frac{3}{5} \times \frac{2}{5} \times \frac{3}{5} \times \frac{1}{5} \times \frac{5}{14}$$

$$= 0.0102$$

compare both probabilities.

$P(\text{yes} | \text{sunny, hot, Normal, Strong}) = 0.00705$

$P(\text{NO} | \text{sunny, hot, Normal, Strong}) = 0.0102$

$\therefore P(\text{NO} | \text{sunny, hot, Normal, Strong})$ having higher probability predicted that play tennis = NO for {sunny, hot, Normal, Strong}.

12 Marks Questions

Q.1 Consider the Given Dataset, Apply Naïve Baye's Algorithm and Predict that if a Tennis Play has following Properties:

- Tennis Play = {Outlook=Sunny, Humidity=High, Wind= Strong}

- Tennis Play = {Outlook=Sunny, Humidity=High, Wind= Weak}

Day	Outlook	Humidity	Wind	Play Tennis
D1	Sunny	High	Weak	No
D2	Sunny	High	Strong	No
D3	Overcast	High	Weak	Yes
D4	Rain	High	Weak	Yes
D5	Rain	Normal	Weak	Yes
D6	Rain	Normal	Strong	No
D7	Overcast	Normal	Strong	Yes
D8	Sunny	High	Weak	No
D9	Sunny	Normal	Weak	Yes
D10	Rain	Normal	Weak	Yes
D11	Sunny	Normal	Strong	Yes
D12	Overcast	High	Strong	Yes
D13	Overcast	Normal	Weak	Yes
D14	Rain	High	Strong	No

→ Firstly, calculate the total prior probability for play tennis (yes/no):

∴ Total count = 14

Play tennis = yes = 9

Play tennis = No = 5

∴ $P(\text{Play tennis} = \text{yes}) = \frac{9}{14} = 0.6428$

$P(\text{Play tennis} = \text{No}) = \frac{5}{14} = 0.3571$

After we need to calculate conditional probability for attributes i.e. outlook, humidity, wind

3) For outlook.

Frequency Table	yes	No
outlook sunny	2	3
overcast	4	0
Rainy	3	2

calculate the probability for sunny=yes and sunny=no

$P(\text{sunny} | \text{yes}) = \frac{P(\text{yes} | \text{sunny}) \cdot P(\text{sunny})}{P(\text{yes})} = \frac{2/5 \times 5/14}{9/14} = \frac{2}{9}$

$P(\text{sunny} | \text{yes}) = 0.2222$

$P(\text{sunny} | \text{No}) = \frac{P(\text{No} | \text{sunny}) \cdot P(\text{sunny})}{P(\text{No})} = \frac{3/5 \times 5/14}{5/14} = \frac{3}{5} = 0.6$

calculate the probability for overcast=yes and overcast=no.

$P(\text{overcast} | \text{yes}) = \frac{P(\text{yes} | \text{overcast}) \cdot P(\text{overcast})}{P(\text{yes})} = \frac{4/4 \times 4/14}{9/14} = \frac{4}{9} = 0.4444$

$P(\text{overcast} | \text{No}) = \frac{P(\text{No} | \text{overcast}) \cdot P(\text{overcast})}{P(\text{No})} = \frac{0/4 \times 4/14}{5/14} = 0$

calculate the probability for rainy=yes and rainy=no.

$P(\text{rainy} | \text{yes}) = \frac{P(\text{yes} | \text{rainy}) \cdot P(\text{rainy})}{P(\text{yes})} = \frac{3/5 \times 5/14}{9/14} = \frac{3}{9} = 0.3333$

$P(\text{rainy} | \text{No}) = \frac{P(\text{No} | \text{rainy}) \cdot P(\text{rainy})}{P(\text{No})} = \frac{2/5 \times 5/14}{5/14} = \frac{2}{5} = 0.4$

3) For Humidity:

Frequency Table	yes	No
Humidity High	3	4
Normal	6	1

calculate prob. for high=yes and high=no.

$$P(\text{High}|\text{yes}) = \frac{P(Y|H) \cdot P(H)}{P(Y)} = \frac{3/14 \times 7/14}{9/14}$$

$$P(H|Y) = 0.3333$$

$$P(\text{High}|\text{No}) = \frac{P(N|H) \cdot P(H)}{P(N)} = \frac{4/14 \times 7/14}{5/14}$$

$$P(H|N) = 0.8$$

calculate prob. for normal=yes and normal=no.

$$P(\text{Normal}|\text{yes}) = \frac{P(Y|N) \cdot P(N)}{P(Y)} = \frac{6/14 \times 7/14}{9/14}$$

$$P(N|Y) = 0.6666$$

$$P(\text{Normal}|\text{No}) = \frac{P(N|N) \cdot P(N)}{P(N)} = \frac{1/14 \times 7/14}{5/14}$$

$$P(N|N) = 0.2$$

3) For Wind:

Frequency Table	yes	No
wind: weak	6	2
strong	3	3

calculate prob. for weak=yes and weak=no

$$P(\text{weak}|\text{yes}) = \frac{P(Y|W) \cdot P(W)}{P(Y)} = \frac{6/14 \times 8/14}{9/14}$$

$$P(W|Y) = 0.6666$$

$$P(\text{weak}|\text{No}) = \frac{P(N|W) \cdot P(W)}{P(N)} = \frac{2/14 \times 8/14}{5/14}$$

$$P(W|N) = 0.4$$

calculate prob. for strong=yes and strong=no

$$P(\text{strong}|\text{yes}) = \frac{P(Y|S) \cdot P(S)}{P(Y)} = \frac{3/14 \times 8/14}{9/14}$$

$$P(S|Y) = 0.3333$$

$$P(\text{strong}|\text{No}) = \frac{P(N|S) \cdot P(S)}{P(N)} = \frac{3/14 \times 8/14}{5/14}$$

$$P(S|N) = 0.6$$

calculate the probability for yes and no.

③ Play tennis = {outlook: sunny, humidity: high, wind: strong}.

$$P(\text{Play}|\text{yes}) = P(\text{yes}) \cdot P(\text{sunny}|\text{yes}) \cdot P(\text{high}|\text{yes}) \cdot P(\text{strong}|\text{yes})$$

$$= 0.64 \times 0.22 \times 0.33 \times 0.33$$

$$= 0.015$$

$$P(\text{Play}|\text{No}) = P(\text{No}) \cdot P(\text{sunny}|\text{No}) \cdot P(\text{high}|\text{No}) \cdot P(\text{strong}|\text{No})$$

$$= 0.35 \times 0.6 \times 0.8 \times 0.6$$

$$= 0.1008$$

compare both probability i.e.

$P(\text{Play}|\text{yes}) = 0.015$ and $P(\text{Play}|\text{No}) = 0.1008$

$P(\text{Play}|\text{No}) = 0.1008$ having highest probability

→ Predict that play = NO for (sunny, high, strong).

② Play = {outlook: sunny, humidity: high, wind: weak}.

$$P(\text{Play}|\text{yes}) = P(\text{yes}) \cdot P(\text{sunny}|\text{yes}) \cdot P(\text{high}|\text{yes}) \cdot P(\text{weak}|\text{yes})$$

$$= 0.64 \times 0.22 \times 0.33 \times 0.66$$

$$= 0.0306$$

$$P(\text{Play}|\text{No}) = P(\text{No}) \cdot P(\text{sunny}|\text{No}) \cdot P(\text{high}|\text{No}) \cdot P(\text{weak}|\text{No})$$

$$= 0.35 \times 0.6 \times 0.8 \times 0.4$$

$$= 0.0691$$

compare both probability i.e.

$P(\text{Play}|\text{yes}) = 0.0306$ & $P(\text{Play}|\text{No}) = 0.0691$

$P(\text{Play}|\text{No}) = 0.0691$ having highest probability

→ Predict that play = NO for (sunny, high, weak).

Q.2 Given the Training data in the table (Buy Computer Data), Predict the Class of the Following New Example Using Naïve Bayes Classifier and Bayes Optimal Classifier. Buy Computer = {Age = <=30, income = medium, student = yes, Credit-Rating = fair} (Same as Q. 4 of 4 marks)

UNIT - 4

2 Mark Questions:

Q1. Define Linear Regression.

Linear Regression is a statistical and machine learning method used to model the relationship between a **dependent variable (y)** and one or more **independent variables (x)** using a straight line.

Formula:

$$y = w_0 + w_1x + \varepsilon$$

where w_0 is intercept, w_1 is slope, and ε is error term.

Example: Predicting house prices based on area (sq.ft).

Q2. Classify Different Types of Linear Regression.

Linear Regression is mainly classified into two types:

1. Simple Linear Regression:

It models the relationship between a single independent variable **x** and a dependent variable **y**.

$$y = w_0 + w_1x$$

Example: Predicting a student's marks based on study hours.

2. Multiple Linear Regression:

It models the relationship between **two or more independent variables** and a dependent variable.

$$y = w_0 + w_1x_1 + w_2x_2 + \dots + w_nx_n$$

Example: Predicting house prices using area, location, and number of rooms.

Q3. Define the Term Regression with Example.

Regression is a predictive modeling technique that estimates the **relationship between input variables (x)** and a **continuous output variable (y)**. It helps forecast values or trends.

Example: Estimating temperature based on humidity and wind speed using regression analysis.

Q4. Define Bayesian Linear Regression.

Bayesian Linear Regression is an extension of linear regression that treats the **model parameters (weights)** as **random variables** with **prior distributions**. Using **Bayes' theorem**, it updates these priors after observing data to obtain **posterior distributions** for predictions.

Formula:

$$P(w|D) = \frac{P(D|w)P(w)}{P(D)}$$

Example: Predicting stock returns while accounting for uncertainty in model parameters.

Q.1 Discuss Linear Regression with their types. (04 Marks)

&

Q.1 Describe Regression in detail and also describe Linear Regression with their Types. (07 Marks)

Regression is a statistical and machine learning technique used to model and analyze the relationship between a **dependent variable (y)** and one or more **independent variables (x)**. Its main purpose is to **predict continuous outcomes** based on input data and to understand how the dependent variable changes with variations in independent variables.

Concept of Regression:

Regression analysis estimates the functional relationship between variables. The general form of a regression model is:

$$y = f(x) + \varepsilon$$

where $f(x)$ represents the model function and ε is the random error term.

Regression helps in prediction, trend analysis, and forecasting.

Example: Predicting house prices based on size, number of rooms, and location.

Linear Regression:

Linear Regression assumes a **linear relationship** between dependent and independent variables. It is represented as:

$$y = w_0 + w_1x + \varepsilon$$

where w_0 is intercept, w_1 is slope, and ε is error term.

It finds the best-fitting straight line (regression line) by minimizing the **sum of squared errors** between observed and predicted values.

Types of Linear Regression:

1. Simple Linear Regression:

Involves **one independent variable** and models the relationship as a straight line.

$$y = w_0 + w_1x$$

Example: Predicting sales based on advertising budget.

2. Multiple Linear Regression:

Uses **two or more independent variables** to predict the dependent variable.

$$y = w_0 + w_1x_1 + w_2x_2 + \dots + w_nx_n$$

Example: Predicting salary based on education, experience, and age.

UNIT - 5

2 Mark Questions:

Q1. Rewrite the Following Terms:

a) Hypothesis Space

The **Hypothesis Space** refers to the complete set of all possible hypotheses (or models) that a learning algorithm can choose from to explain the given data.

b) Hypothesis Space Search

Hypothesis Space Search is the process of searching through all possible hypotheses in the hypothesis space to find the one that best fits the training data.

Q2. Restate the Decision Tree Algorithm.

A **Decision Tree Algorithm** is a supervised learning method that uses a **tree-like model** to make decisions based on data attributes. It works by **recursively splitting** the dataset into smaller subsets using measures like **Information Gain** or **Gini Index** to select the best attribute at each node.

The process continues until all instances are classified or a stopping criterion (like purity or depth limit) is met.

Q3. Define the Following Terminology w.r.t. Decision Tree:

a) Root Node:

The topmost node in a decision tree that represents the **entire dataset**. The root node is chosen based on the **attribute with the highest information gain** and is the starting point for all splits.

b) Leaf Node:

The **terminal node** that gives the **final output or class label**. No further splitting occurs at this point.

c) Splitting:

The process of **dividing a dataset** into smaller subsets based on an attribute's values to create purer nodes and improve classification accuracy.

d) Pruning:

A technique used to **remove unnecessary branches** from a fully grown decision tree. It helps to **reduce overfitting** and improve model generalization on unseen data.

Q4. Rewrite the Term Hypothesis Space Search with Example.

Hypothesis Space Search refers to the method by which a learning algorithm explores the set of all possible models (hypotheses) to identify the one that best fits the training data.

In **Decision Tree Learning**, the hypothesis space consists of all possible tree structures. The algorithm uses a **greedy search** strategy, selecting at each step the attribute that provides the **highest information gain** to create an effective tree.

Example: The **ID3 algorithm** searches by repeatedly choosing the best attribute until it forms a complete tree.

Q5. Write a Short Note on Decision Tree.

A **Decision Tree** is a **supervised learning algorithm** used for both **classification and regression tasks**. It models decisions using a **tree structure**, where each internal node tests an attribute, each branch represents a decision outcome, and each leaf node provides a final class label or prediction.

It splits data recursively using metrics like **Information Gain** or **Gini Index** and can handle both categorical and numerical data. Decision trees are **easy to interpret, non-parametric**, and often serve as the foundation for ensemble methods like **Random Forests**.

Example: Predicting whether a loan application will be approved based on income, credit history, and age.

Q6. State the Following Terms in the Context of Decision Tree:

a) Hypothesis Space:

In a Decision Tree, the **hypothesis space** is the set of all possible trees that can be formed using the available attributes. Each possible tree represents one hypothesis or classification rule.

b) Hypothesis Space Search:

The process by which the Decision Tree algorithm **navigates through the hypothesis space** to find the most accurate tree. It uses a **greedy search approach**, selecting attributes that yield the **best split** at each step.

Example: The **ID3 algorithm** performs a top-down greedy search through the hypothesis space.

Q7. Give Any Example of Hypothesis Space Search.

Example:

In the **ID3 Decision Tree Algorithm**, the system performs a **hypothesis space search** by starting with the entire dataset and selecting the attribute that provides the **maximum information gain**.

At each level, the algorithm chooses the next best attribute to split the data until pure or homogeneous subsets are formed. This process explores the hypothesis space to build the best-fitting tree structure for classification.

4 Mark Questions:

Q1. Discuss the Hypothesis Space Search in Decision Tree in Detail.

In a **Decision Tree**, the **hypothesis space** consists of all possible trees that can be formed using available attributes. The **Hypothesis Space Search** is the process of exploring this set to find the tree that best fits the training data.

Decision Tree algorithms, such as **ID3** and **C4.5**, use a **greedy top-down search** approach, beginning with the full dataset and selecting the attribute that offers the **highest information gain** to split the data.

This search continues recursively on each branch until all data are perfectly classified or a stopping condition is reached. Although this method doesn't guarantee a globally optimal solution, it efficiently finds a **locally optimal tree** that provides good classification accuracy.

Example: In ID3, attributes are selected based on maximum Information Gain at each node to build an effective decision tree.

Q2. Discuss the Following Terminology w.r.t. Decision Tree.

1. **Root Node:**

The topmost node representing the entire dataset. It is selected based on the attribute that gives the **maximum information gain** and initiates the tree construction.

2. **Leaf Node:**

The terminal node that provides the **final class label or output**. No further division occurs at this stage.

3. **Splitting:**

The process of **dividing data** into subsets based on attribute values. Splitting aims to increase the homogeneity of nodes.

4. **Branch/Sub-Tree:**

Each outcome of a split forms a **branch** leading to another node. A **sub-tree** is a smaller section of the overall tree structure.

5. **Pruning:**

The process of **removing unnecessary branches** after tree construction to avoid overfitting and improve generalization.

6. **Parent/Child Node:**

The node that splits into sub-nodes is the **parent**, while the resulting nodes are called **child nodes**.

Q3. Illustrate the Different Methods of Sampling.

Sampling is a statistical process of selecting a subset of data (sample) from a larger dataset (population) to estimate population characteristics efficiently. In machine learning, sampling helps in model training and hypothesis evaluation.

The main methods of sampling are:

1. **Random Sampling:** Every element has an equal chance of selection.
2. **Stratified Sampling:** Population is divided into groups (strata) based on specific features, and samples are drawn proportionally.
3. **Systematic Sampling:** Elements are chosen at regular intervals (e.g., every 5th observation).
4. **Cluster Sampling:** Population is divided into clusters, and entire clusters are selected randomly.

Example: Selecting every 10th data record for testing in a large dataset.

These methods ensure accurate hypothesis evaluation while reducing computational cost.

Q4. Discuss and List the Different Types of Sampling.

Sampling is the process of selecting a subset (sample) from a larger dataset (population) to analyze and make predictions efficiently. In **Machine Learning**, sampling is used to divide data into training and testing sets for performance evaluation.

The main types of **Probability Sampling** are:

1. **Random Sampling:**
Every element in the dataset has an equal chance of being selected.
Example: Selecting 100 data points randomly from a dataset of 1,000 records.
 2. **Stratified Sampling:**
The population is divided into groups (strata) based on specific attributes, and samples are drawn proportionally from each stratum.
Example: Dividing students by grade level before sampling.
 3. **Systematic Sampling:**
Samples are selected at regular intervals (every k th record) from an ordered dataset.
Example: Choosing every 5th data entry from a database.
 4. **Cluster Sampling:**
The population is divided into clusters, and entire clusters are randomly selected for analysis.
Example: Selecting a few random schools from a district for survey analysis.
-

7 Marks Questions:

Q1. Illustrate in Detail the Concept of Decision Tree and also Outline the Main Steps Followed in Constructing a Decision Tree Algorithm.

A **Decision Tree** is a **supervised learning algorithm** used for **classification and regression tasks**. It models decisions and their possible outcomes using a **tree-like structure** of nodes and branches.

Each **internal node** represents a test on an attribute, each **branch** corresponds to the outcome of the test, and each **leaf node** represents a class label or output value.

The Decision Tree works by splitting data recursively based on attribute values, aiming to create subsets that are as pure (homogeneous) as possible.

Advantages:

- Easy to interpret and visualize.
- Handles both categorical and numerical data.
- Requires little data preprocessing.

Example:

A decision tree predicting whether a student passes or fails based on study hours and attendance.

Steps in Constructing a Decision Tree Algorithm:

1. **Start with the Entire Dataset:**
Consider all attributes and instances.
 2. **Select the Best Attribute:**
Use criteria such as **Information Gain** or **Gini Index** to determine the attribute that best splits the data.
 3. **Split the Dataset:**
Partition the data into subsets based on the selected attribute's values.
 4. **Repeat Recursively:**
Continue the process for each subset until all instances in a node belong to the same class or other stopping criteria are met.
 5. **Prune the Tree:**
Remove unnecessary branches to simplify the model and reduce overfitting.
-

Q2. Examine in Detail about Hypothesis Testing with their Types.

Hypothesis Testing is a statistical method used to make inferences or decisions about a population based on sample data. It helps determine whether an observed result is due to chance or a significant underlying effect.

A **hypothesis** is an assumption made about a population parameter, which is then tested using sample data and probability theory.

Steps in Hypothesis Testing:

1. **Formulate Hypotheses:**

- **Null Hypothesis (H_0):** Assumes no significant difference or effect.
- **Alternative Hypothesis (H_1):** Assumes there is a significant difference or effect.

2. **Set the Significance Level (α):**

Common values are 0.05 or 0.01, defining the probability of rejecting H_0 when it is true.

3. **Select the Appropriate Test:**

Choose a test statistic such as z , t , or χ^2 depending on data type and sample size.

4. **Compute the Test Statistic:**

Calculate the value using sample data to measure the degree of deviation from H_0 .

5. **Make a Decision:**

Compare the calculated value with the critical value or p-value.

- If $p < \alpha$, reject H_0 .
- If $p \geq \alpha$, fail to reject H_0 .

Types of Hypothesis Testing:

1. **One-Tailed Test:**

Tests whether a parameter is either greater or less than a specific value.

2. **Two-Tailed Test:**

Tests whether a parameter is significantly different from a specific value (either higher or lower).

Example:

Testing whether a new medicine reduces blood pressure compared to an existing one using a t-test.

Q3. Illustrate Basic Sampling Theory with Their Step-by-Step Process.

Sampling Theory deals with selecting a representative subset (sample) from a population to estimate population characteristics. It provides methods to ensure that conclusions drawn from the sample are statistically valid for the entire population.

Sampling is essential in **machine learning** for dividing data into **training and testing sets** to evaluate model performance efficiently.

Step-by-Step Process of Sampling:

1. **Define the Population:**

Identify the complete set of data or elements you want to study.

2. **Determine the Sampling Frame:**

Specify the list or source from which samples will be drawn (e.g., database, survey list).

3. **Select the Sampling Method:**

Choose a method such as **Random, Stratified, Systematic, or Cluster Sampling** depending on data characteristics.

4. **Decide the Sample Size:**

The sample should be large enough to represent the population accurately while minimizing cost and time.

5. **Select the Sample:**

Draw the sample using the chosen technique ensuring fairness and lack of bias.

6. **Collect and Analyze Data:**

Gather observations from the selected sample and perform statistical analysis to estimate population parameters.

7. **Evaluate Sampling Error:**

Compare sample results with actual population values to estimate sampling error and reliability.

Example:

Selecting 100 students randomly from a college of 1,000 to study average marks.

Q4. Analyse the Concept of Basic Sampling Theory with the Step-by-Step Process Involved in Creating One with a Neat Diagram.

Sampling Theory is the process of selecting a subset (sample) from a larger population to represent the whole group. In **Machine Learning**, sampling helps in selecting data for **training and testing** while maintaining representativeness of the original dataset. It ensures that the sample accurately reflects the population and minimizes bias or errors.

Steps in Sampling Process:

Step 1 – Identifying the Population Set: Define the complete group or dataset you are interested in.

Example: All students in a college.

Step 2 – Determination of the Size of the Sample Set: Decide how many data points (sample size) you will take from the population.

This is called the sample size.

Step 3 – Providing a Basis for Selection of Samples: Determine the criteria or medium for choosing samples. Decide **how** you will pick them (method of selection).

Step 4 – Picking Out Samples Using Techniques: Select actual data using appropriate sampling methods such as:

1. **Simple Random Sampling** – pick randomly.
2. **Systematic Sampling** – pick every n th element.
3. **Stratified Sampling** – pick proportionally from groups.

Step 5 – Checking Representativeness:

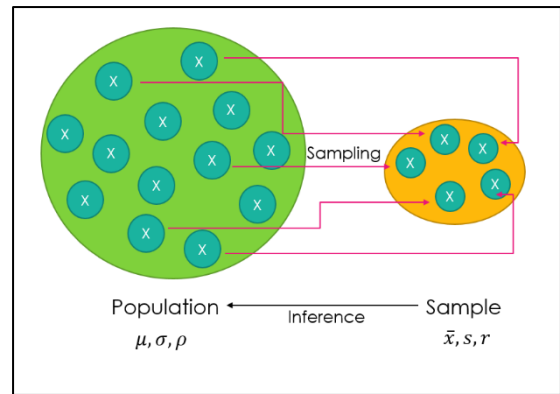
Ensure that the selected sample truly represents the population (e.g., gender, marks distribution) without major bias or differences.

Step 6 – Checking for Errors or Inaccuracies:

Verify whether the sample introduces bias or mistakes that could lead to wrong conclusions.

Step 7 – Final Sample Set: After verification, choose the final subset of data.

This **final sample** is used for **analysis or training** in Machine Learning.



Q5. Discuss the Concept of Hypothesis Testing by Explaining Its Types and Graphical Representation with Suitable Examples.

Hypothesis Testing is a form of statistical inference used to draw conclusions about a population parameter using data from a sample. It helps in deciding whether a claim or assumption about a population is true or false based on sample evidence.

It compares two opposite ideas — one assuming *no effect or difference* (**Null Hypothesis**) and another assuming *an effect or difference* (**Alternative Hypothesis**). We collect and analyze sample data to decide which hypothesis is more likely to be true.

Example:

If a company claims its website gets **50 visitors per day**, hypothesis testing helps verify whether this claim is true or if the actual average is different.

Defining Hypotheses:

1. Null Hypothesis (H_0):

It is the default assumption that there is *no significant difference* or effect.

Example:

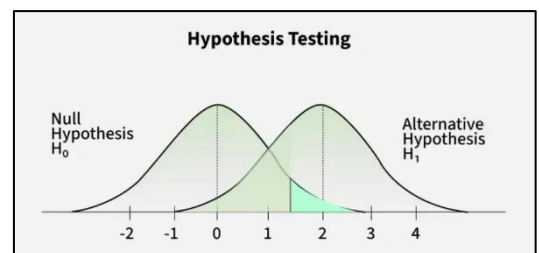
$H_0 : \mu = 50$ (The average daily visitors are 50.)

2. Alternative Hypothesis (H_1):

It is the logical opposite of the null hypothesis, suggesting a difference or effect exists.

Example:

$H_1 : \mu \neq 50$ (The average daily visitors are not 50.)



Types of Hypothesis Testing:

1. One-Tailed Test:

Used when the change is expected in only one direction (either increase or decrease).

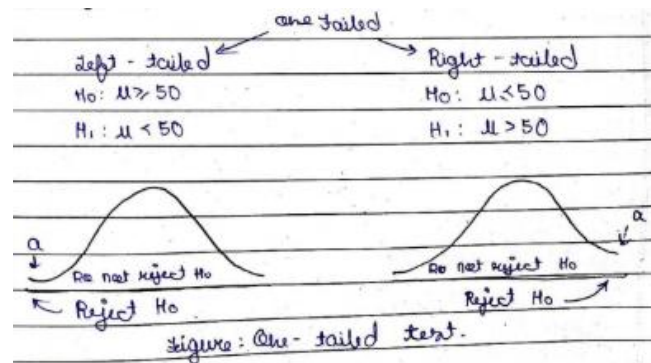
- **Left-Tailed Test:** Checks if the value is *less than expected*.

Example: $H_0 : \mu \geq 50, H_1 : \mu < 50$

- **Right-Tailed Test:** Checks if the value is *greater than expected*.

Example: $H_0 : \mu \leq 50, H_1 : \mu > 50$

Example: Testing if a new algorithm increases model accuracy (only upward change is tested).



2. Two-Tailed Test:

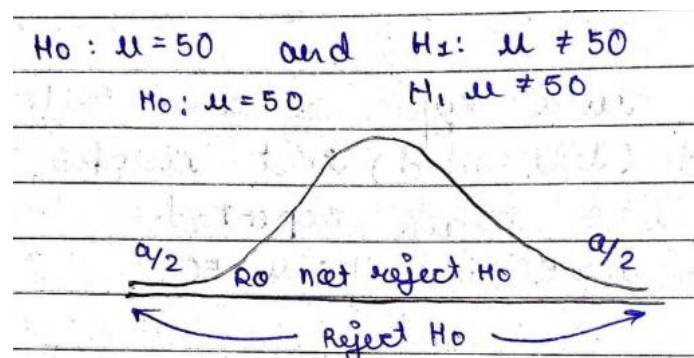
Used when the difference can occur in *either direction* — higher or lower than the expected value.

Example:

$$H_0 : \mu = 50$$

$$H_1 : \mu \neq 50$$

It tests whether a parameter differs significantly from the hypothesized mean, regardless of direction.



12 Marks Questions:

Q1. Rewrite the Concept of Basic Sampling Theory and Also Rewrite the Following About Sampling:

a) Process of Sampling

b) Types of Sampling

Sampling Theory is a statistical concept that deals with selecting a subset (sample) from a large group (population) to analyze and draw conclusions about the entire population.

The main objective of sampling is to obtain a **representative sample** that reflects the true characteristics of the population without bias.

In **Machine Learning**, sampling is widely used for dividing data into **training and testing sets** to evaluate model performance effectively.

a) Process of Sampling:

The sampling process involves systematic steps to ensure that the selected sample represents the population accurately.

The steps are as follows:

Step 1 – Identifying the Population Set:

Define the complete group of elements or individuals under study.

Example: All students in a college.

Step 2 – Determination of Sample Size:

Decide how many data points (samples) to select from the population. This is called **sample size**.

Step 3 – Providing a Basis for Selection:

Determine the medium or basis for selecting the samples — i.e., how samples will be chosen.

Step 4 – Selecting the Samples Using Techniques:

Choose actual data using appropriate sampling techniques such as:

- **Simple Random Sampling** – selecting data points randomly.
- **Systematic Sampling** – picking every n th element.
- **Stratified Sampling** – selecting proportionally from groups.

Step 5 – Checking Representativeness:

Ensure the chosen sample reflects population characteristics like gender, age, or marks distribution without bias.

Step 6 – Checking for Errors or Inaccuracies:

Verify that no sampling bias or errors have been introduced.

Step 7 – Finalizing the Sample Set:

After verification, the final subset is selected for analysis or training in Machine Learning.

b) Types of Sampling:

Sampling techniques are mainly categorized under **Probability Sampling**, where every element in the population has a known and non-zero chance of being selected.

The main types are:

1. Random Sampling:

Each element has an equal chance of being chosen.

Example: Selecting 100 students randomly from a class of 1,000.

2. Systematic Sampling:

Samples are selected at regular intervals (e.g., every 5th record).

Example: Choosing every 10th customer from a sales list.

3. Stratified Sampling:

The population is divided into subgroups (strata) based on common characteristics, and samples are drawn proportionally from each stratum.

Example: Sampling students from each year (1st, 2nd, 3rd, and 4th).

4. Cluster Sampling:

The population is divided into clusters, and entire clusters are randomly selected.

Example: Selecting a few schools randomly to study student performance.
